Systems Privacy Project 2

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1 Introduction

The existing implementation of the Tor path selection algorithm only prevents nodes within the same /16 subnets from appearing in a single circuit. While, this offers some degree of protection, it might not be enough when dealing with an adversary who can perform traffic correlation at country level.

This project focuses on implementing a new Tor path selection algorithm that alters the weighting mechanism to reduce the likelihood of selecting nodes within the same country.

Finally, we compared both implementations and conducted testing with JUnit to assess both reliability and performance of both solutions.

2 Implementation

2.1 Consensus

In order to pass the *consensus* file effectively we used a BufferedReader with a 16KB buffer. We perform 2 parses over the data. First, we parse line by line looking for the correct line prefix to retrieve the correspondent information, and we use it to populate the Node object of every thing except its country. Then in a second parsing we perform geolocation of the nodes. This was done this way to take advantage of batching IP addresses into one HTTP request, in order to achieve a faster parsing.

We use IPquery.io[1] batch API endpoint which accepts a comma-separated list of addresses. We chose this solution since it is a hassle-free solution based on a API, which means no library integrations or local databases, and the batching allows up to 100000 addresses per request[2], which we leverage to get all of the IP addresses necessary for a full *consensus* document in no more than two batches.

2.2 Current Path Selection (Algorithm 1)

This first algorithm attempts to replicate the current Tor path selection algorithm. To be more realistic, when the application bootstraps it checks the persistent

[1] https://ipquery.io/

[2] https://ipquery.gitbook.io/ipquery-docs

state to determine whether a guard set is already stored. If so, that guard set is reused.

Node families that are not present in the *consensus* are obtained from the most recent server descriptors available at the time of execution. To parse these descriptors, we use the Tor Metrics library [3].

2.3 Geography-Aware Path Selection (Algorithm 2)

This second algorithm tries to address the limitation in Tor's path selection referred in the Introduction by extending the design used above. We use a weighted selection approach that focuses on limiting the potential of a circuit to exist inside a single country which could expose the users to traffic correlation attacks. We introduce two additional security parameters α and β which are used during middle node and guard node selection, respectively.

For middle node selection, we assign a different increasing bonus based on a geographical diversity score c, where c is the number of different countries in the circuit. Then we change the weight of a candidate by multiplying its bandwidth weight by $1 + \beta \times c$.

For guard node selection, we assign a bonus based on if a candidate's country is present in the circuit, which only contains the exit node at this point. If not, the guard candidate's country is not present in the circuit's existing country set (which contains only the exit node at this stage), its bandwidth weight receives a bonus of $1 + \alpha$.

This design incentivizes the choice of nodes in separate countries while keeping the same probabilistic approach offered in traditional path selection.

2.4 Results

In order to compare the performance and security of the path selection implementations, we analyse both the size (cardinality) and the *Shannon* entropy of each node set (global, guard, middle and exit), and we compare the algorithms in terms of circuit bandwidth.

We conducted two experiments using 10000 IP addresses and the most recent consensus available at the time of execution, varying the parameters α and β to analyze their impact. The first experiment uses moderate parameter values (α = 0.5 and β = 0.5), which we treat as the default. In the second experiment, we choose a higher α (α = 0.8), since this parameter primarily influences guard selection and is therefore relevant for mitigating the risk of end-to-end correlation attacks, and a much smaller β (β = 0.2) because there is comparatively less risk inherent in the choice of middle nodes. The results of the first experiment are presented in Table 1 and 2 and in Figure 1. The results of the second experiment are presented in Table 3 and 4 and in Figure 2.

[3] https://metrics.torproject.org/metrics-lib/

set	size	entropy
Global	4910.0	9.0135
Guard	3.0	1.0872
Middle	4208.0	11.6601
Exit	2111.0	10.3856

Table 1: Results Current Path Selection (α = 0.5, β = 0.5)

set	size	entropy
Global	4872.0	9.0112
Guard	3.0	1.1067
Middle	4219.0	11.6795
Exit	2075.0	10.3636

Table 2: Results Geography-Aware Path Selection (α = 0.5, β = 0.5)

The first experiment shows identical set sizes and entropy values for both implementations. As expected, only three guard nodes are chosen across all circuits, because the guard set is persisted at the beginning and the path selection always chooses one of these guards. With this evaluation, we do not observe substantial differences between the two algorithms.

A critical observation is that these metrics may not be well suited to highlight the differences in which we are interested. The set size does not convey any information about the geographical distribution of nodes, and the entropy is computed over nodes rather than over countries. Nodes in the same circuit are likely to belong to different countries, but this evaluation may not capture that aspect. To investigate this aspect in more detail, we developed a test suite that includes an experiment that compares country diversity within circuits for the two algorithms over more than 200 rounds. Although, due to the probabilistic nature of the algorithms, the results may vary, Algorithm 2 usually achieves better country diversity than Algorithm 1.

set	size	entropy
Global	4914.0	8.9940
Guard	3.0	1.0731
Middle	4238.0	11.6814
Exit	2062.0	10.3362

Table 3: Results Current Path Selection (α = 0.8, β = 0.2)

set	size	entropy
Global	4910.0	8.9947
Guard	3.0	1.0819
Middle	4244.0	11.6786
Exit	2050.0	10.3272

Table 4: Results Geography-Aware Path Selection (α = 0.8, β = 0.2)

In the second experiment, the results are again not very conclusive, and the metrics do not clearly expose the impact of the geographic-aware path selection parameters.

Regarding circuit bandwidth, since the total number of constructed circuits is quite large, in Figure 1 and Figure 2 we restrict the plot to the first 40 circuits as a representative sample. Each pair of bars shows the minimum bandwidth of the same circuit identifier under Algorithm 1 and Algorithm 2. We can see that the two algorithms achieve comparable bandwidths across most circuits, with neither consistently dominating the other. In conclusion, despite the second algorithm having more restrictions those did not affect the performance when compared to Algorithm 1.

A Images

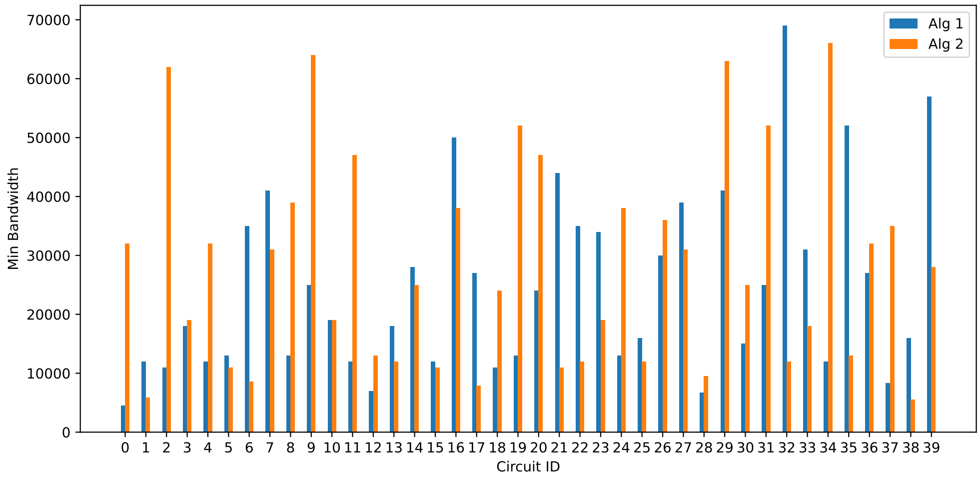

Figure 1: Bandwidth comparison

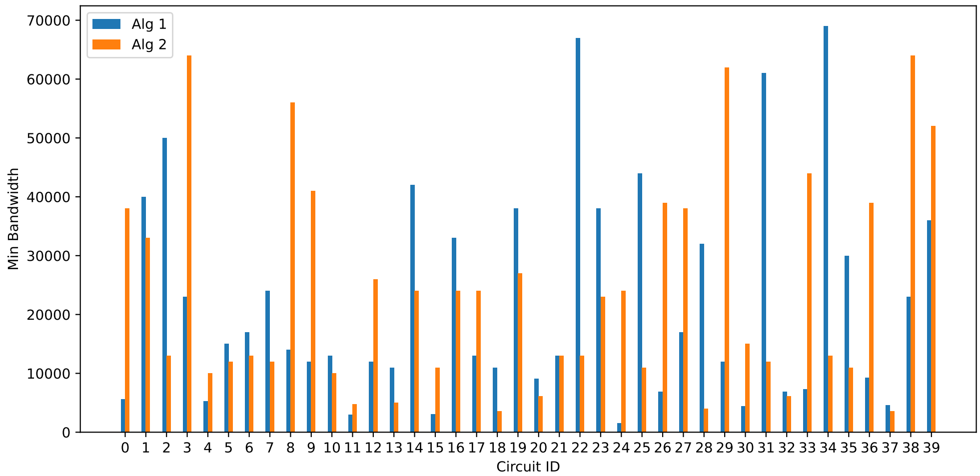

Figure 2: Bandwidth comparison